

ELAZIG, Turkey

WMO: 172020

Lat: 38.606N

Lon: 39.297E

Elev: 881

StdP: 91.18

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.0	-7.8	-16.3	1.0	-5.9	-14.1	1.2	-4.1	9.0	4.8	8.0	3.8	1.9	250	0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	38.2	18.5	37.0	18.1	35.8	17.8	19.9	35.6	19.1	34.5	18.5	33.6	3.3	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.2	11.2	22.8	13.2	10.5	23.3	12.2	9.9	23.3	60.7	35.4	58.1	34.5	55.9	33.6	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	6.7	5.7	-13.0	40.4	3.5	1.3	-15.5	41.3	-17.6	42.1	-19.5	42.8	-22.1	43.7		
(5)			-13.7	21.2	3.4	1.5	-16.1	22.2	-18.1	23.1	-20.0	23.9	-22.5	25.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	0.7	2.2	7.0	12.2	17.2	22.7	27.1	27.1	21.7	15.3	7.7	2.5		
	DBStd	9.80	3.72	3.98	3.59	3.49	2.95	2.73	2.10	2.01	2.90	3.29	3.42	4.03		
	HDD10.0	944	288	219	104	16	0	0	0	0	0	3	81	233		
	HDD18.3	2512	547	453	350	186	56	2	0	0	7	102	319	491		
	CDD10.0	2288	0	0	12	82	224	381	528	529	351	168	12	0		
	CDD18.3	813	0	0	0	2	22	133	270	271	108	9	0	0		
	CDH23.3	11819	0	0	1	48	429	1953	3904	3809	1499	176	0	0		
	CDH26.7	6428	0	0	0	7	109	953	2340	2292	697	30	0	0		
(14) Wind	WSAvg	2.4	2.1	2.2	2.6	2.5	2.5	2.7	2.8	2.5	2.5	2.2	2.1	2.1		
(15) Precipitation	PrecAvg	456	48	55	54	68	50	11	4	1	10	50	53	52		
	PrecMax	624	109	108	128	114	140	29	16	6	33	136	97	118		
	PrecMin	274	4	22	2	22	10	0	1	0	0	6	3	0		
	PrecStd	85	22	22	35	26	33	9	4	1	10	32	28	31		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.3	15.1	20.8	27.4	32.0	36.9	40.2	40.0	36.1	29.4	19.6	12.8		
		MCWB	5.7	7.8	10.2	13.7	16.9	17.6	19.3	18.6	17.0	15.0	10.5	7.8		
	2%	DB	8.7	12.3	18.5	24.5	29.4	35.0	38.8	38.3	34.2	27.2	17.8	10.8		
		MCWB	4.8	6.4	9.3	12.7	15.2	17.6	18.5	18.6	16.8	14.2	9.6	6.9		
	5%	DB	7.5	10.2	16.7	22.3	27.8	33.2	37.2	37.2	32.7	25.2	15.9	9.4		
		MCWB	4.4	5.2	8.4	11.7	14.7	16.8	18.1	18.3	16.4	13.4	8.9	6.1		
	10%	DB	6.2	8.5	14.3	20.1	25.9	31.9	35.9	36.0	30.9	23.2	14.1	8.2		
		MCWB	3.5	4.5	7.3	10.9	14.2	16.2	17.9	18.0	15.7	12.8	8.3	5.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.0	9.1	11.5	15.1	18.4	19.9	21.1	21.0	18.9	16.2	12.1	9.2		
		MCDB	8.9	12.8	18.1	23.3	29.0	32.8	37.9	36.7	33.6	25.0	16.7	11.6		
	2%	WB	5.6	7.2	10.1	13.5	16.7	18.6	20.0	19.9	17.8	15.2	10.9	7.7		
		MCDB	7.7	10.9	16.5	21.9	25.5	32.2	35.9	36.0	32.1	23.8	15.2	9.9		
	5%	WB	4.7	6.0	9.0	12.7	15.8	17.7	19.2	19.2	16.9	14.4	9.9	6.7		
		MCDB	6.8	9.2	15.0	20.4	24.7	31.0	34.8	35.0	30.6	22.7	14.1	8.8		
	10%	WB	3.9	4.9	8.1	11.7	14.8	16.9	18.5	18.5	16.1	13.5	9.0	5.7		
		MCDB	5.9	7.8	13.2	18.8	23.6	29.7	33.9	34.2	29.4	21.4	13.3	7.7		
(35) Mean Daily Temperature Range	MDBR		7.9	9.7	11.7	13.1	14.9	17.1	17.3	17.6	17.3	14.1	11.5	7.6		
	5% DB	MCDBR	9.0	13.1	15.5	17.0	18.3	18.8	19.3	19.3	19.5	17.3	14.5	9.2		
		MCWBR	5.9	8.2	8.1	8.0	7.5	7.2	7.7	7.9	8.7	8.0	8.0	5.8		
	5% WB	MCDBR	7.1	10.9	13.0	14.5	16.1	17.9	17.6	18.1	18.2	14.9	11.7	7.0		
		MCWBR	5.0	7.1	7.2	7.4	7.3	7.3	7.4	8.1	8.5	7.2	6.9	4.7		
(40) Clear-Sky Solar Irradiance	taub		0.331	0.365	0.403	0.452	0.419	0.366	0.362	0.372	0.341	0.386	0.343	0.329		
	taud		2.245	2.175	2.129	2.029	2.193	2.378	2.398	2.350	2.422	2.246	2.336	2.309		
	Ebn at Noon		827	844	849	827	863	909	910	893	905	811	810	802		
	Edh at Noon		100	123	142	167	145	120	117	120	105	113	91	88		
(44) All-Sky Solar Radiation	RadAvg		2.00	2.96	4.14	5.19	6.50	7.95	8.05	7.14	5.83	3.86	2.62	1.82		
	RadStd		0.32	0.46	0.38	0.55	0.43	0.36	0.21	0.22	0.23	0.29	0.31	0.39		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+1.34	N/A	-0.73	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)	N/A	N/A	N/A	N/A	-0.87	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page